Last Time

Dependent Samples

Advantages to dependent sample designs

Effect Sizes for Mean Comparisons
- Cohen's d
- Hedges' g
- Glass' Δ

Power
- Factors affecting power

Today

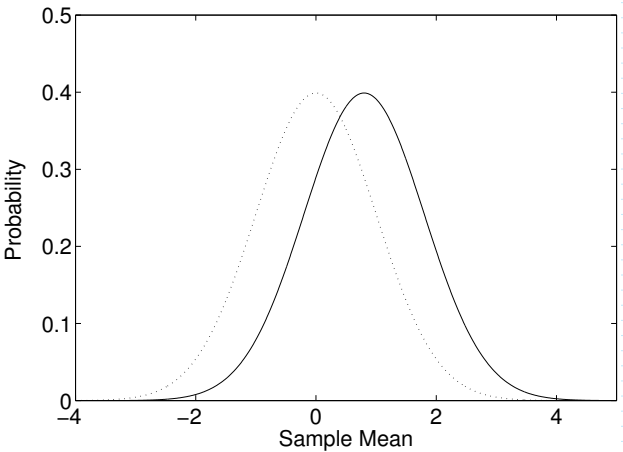
Finishing Effect Sizes

Power
- Factors affecting power
- Power Computation

Examples
- The Noncentrality Parameter

Determining Sample Size

Effect Size



Cohen's d

Z-test

$$d = \left| \frac{\overline{X} - \mu_0}{\sigma} \right|$$

T-test

$$d = \left| \frac{\overline{X} - \mu_0}{s} \right|$$

Independent Samples T-test

$$d = \left| \frac{\overline{X}_1 - \overline{X}_2}{s_P} \right|$$

Cohen's Convention

d	Size
.2	Small
.5	Medium
.8	Large

GSS Example I

```
> library('effectsize')
> t.test(money,mu=12513)

 One Sample t-test

data:  money
t = 13.031, df = 872, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 12513
95 percent confidence interval:
 21548.13 24752.30
sample estimates:
mean of x
 23150.21

> cohens_d(money,mu=12513)
Cohen's d |        95% CI
-------------------------
0.44      | [0.37, 0.51]

- Deviation from a difference of 12513.
```

GSS Example II

```
> t.test(money~happy,alternative="less")

 Welch Two Sample t-test

data:  money by happy
t = -3.0283, df = 161.28, p-value = 0.001432
alternative hypothesis: true difference in means between group 0 and group 1 is less than 0
95 percent confidence interval:
      -Inf -2687.825
sample estimates:
mean in group 0 mean in group 1
       17924.87        23849.19

 > cohens_d(money~happy,alternative="less",pooled_sd=FALSE)
Cohen's d |         95% CI
-------------------------
-0.28     | [-Inf, -0.12]

- Estimated using un-pooled SD.
- One-sided CIs: lower bound fixed at [-Inf].
```

Power

State of the World	Decision	
	H_0 True	H_0 False
H_0 True	1 - α	α
H_0 False	β	1-β

Power

State of the World	Decision	
	H_0 True	H_0 False
H_0 True	1 - α	α
H_0 False	β	1-β

Define significance level:

$$\alpha = Pr(\text{Type } I \text{ error}) = Pr(\text{Reject } H_0 | H_0 \text{ true})$$

Power

State of the World	Decision	
	H_0 True	H_0 False
H_0 True	1 - α	α
H_0 False	β	1-β

Define significance level:

$$\alpha = Pr(\text{Type } I \text{ error}) = Pr(\text{Reject } H_0 | H_0 \text{ true})$$

Define power:

$$\begin{aligned} 1-\beta &= 1 - Pr(\text{Type } II \text{ error}) \\ &= Pr(\text{Reject } H_0 \,|\, H_0 \text{ false}) \end{aligned}$$

Error Probabilities

- The probability of rejecting a true H_0 is α. We set this probability directly when we select a significance level.
- The probability of rejecting a false H_0 depends on the effect size! If we specify an effect size, we can compute this probability, which is called *power* and equals $1 - \beta$

Error Probabilities

- The probability of rejecting a true H_0 is α. We set this probability directly when we select a significance level.
- The probability of rejecting a false H_0 depends on the effect size! If we specify an effect size, we can compute this probability, which is called *power* and equals $1 - \beta$
  Power is the ability of a test to detect/reject a false null hypothesis.
  We can only compute $1 - \beta$ if we know H_1 - if we specify an effect size.
  Typically we don't know H_1

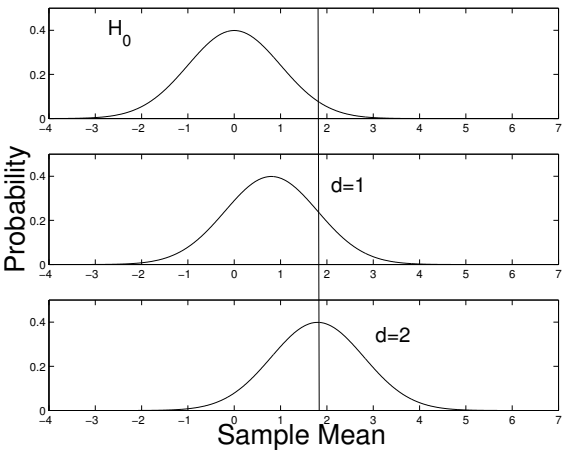
Power and Significance

- Significance levels α are selected in advance, based on practical considerations.
- Issues of power are often not considered until after a failure to reject H_0. However, there are ways to insure good power, which should be used *before* testing H_0.

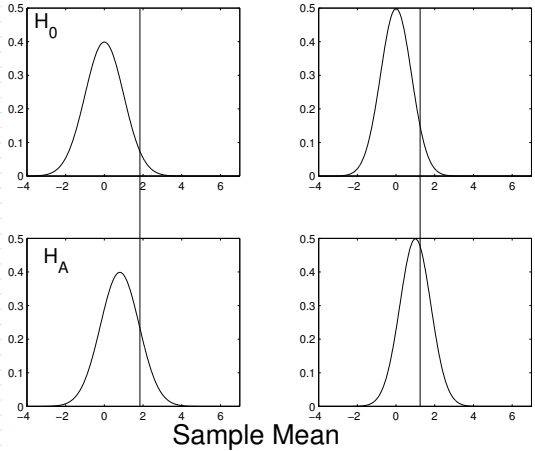
Factors Affecting Power

1. Effect Size (Cohen's d)
2. Sample Size
3. Significance level
   - One vs. two-tailed tests

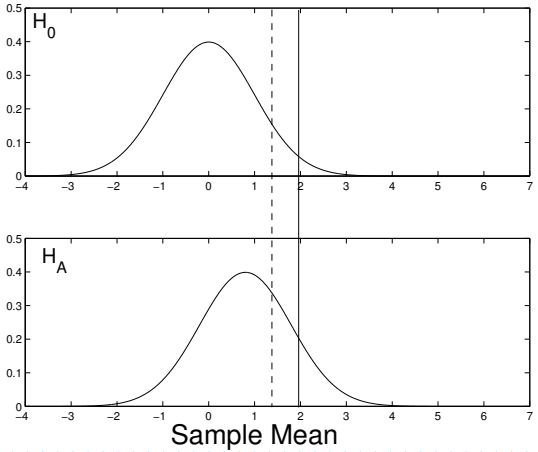
Effect Size



Sample Size



Alpha Level



Power and Standard Error

Another way to increase power: Reduce the variance $\sigma_{\overline{X}_1 - \overline{X}_2}$.

Power and Standard Error

Another way to increase power: Reduce the variance $\sigma_{\overline{X}_1 - \overline{X}_2}$.

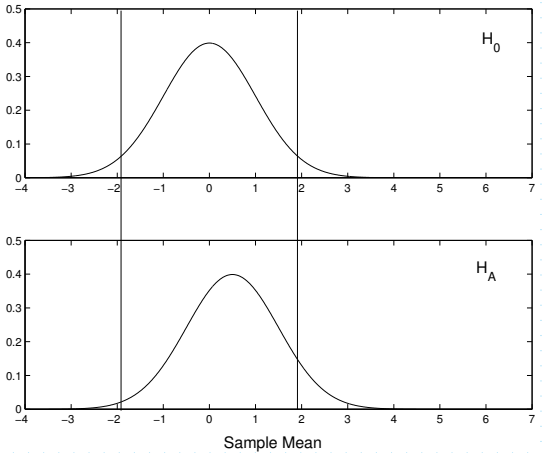
- Dependent samples

Summary

Power increases with

- Larger effect sizes
- Larger sample sizes
- Larger α-levels (one-tailed tests)
- Dependent samples designs

A Note on Two-Tailed Tests



Today's Plan

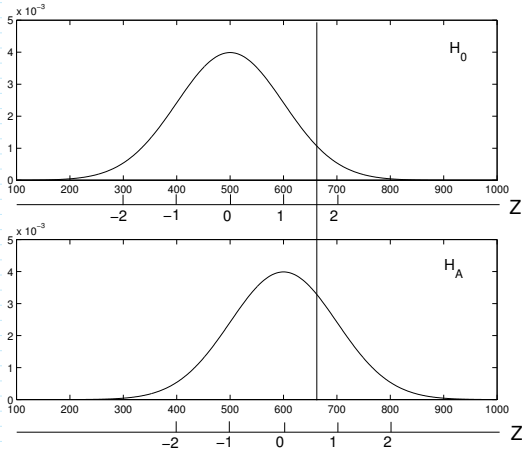
- Go through z-test power step by step
- Relate power to effect size for t-tests
- Demonstrate using R for power computations
- Determining sample size for desired power and effect size

Power Computations

Remember: We need a value for μ_1 (effect size).

$$H_0: \quad \mu = \mu_0$$
$$H_1: \quad \mu = \mu_1$$

Setup


x 10^-3
5
4
3
2
1
0
100
200
300
400
500
600
700
800
900
1000
H0
−2
−1
0
1
2
Z
x 10^-3
5
4
3
2
1
0
100
200
300
400
500
600
700
800
900
1000
HA
−2
−1
0
1
2
Z

Example

In a rat, the number of hippocampal neurons of a certain type has mean $\mu = 500$ and variance $\sigma^2 = 1000^2$. 100 rats are treated with a memory-enhancing drug. On average, their hippocampi contain more of these cells than for untreated rats.
If we do a z-test ($\alpha = .05$), what is the power of this test against the alternative that the population mean for treated rats is really 600 cells?

Example

In a rat, the number of hippocampal neurons of a certain type has mean $\mu = 500$ and variance $\sigma^2 = 1000^2$. 100 rats are treated with a memory-enhancing drug. On average, their hippocampi contain more of these cells than for untreated rats.
If we do a z-test ($\alpha = .05$), what is the power of this test against the alternative that the population mean for treated rats is really 600 cells?
The desired effect size is $\frac{600 - 500}{1000} = 0.10$.

Problem

- Hypotheses:

$$H_0: \quad \mu = 500$$
$$H_1: \quad \mu = 600$$

- What value of $\bar{X}$ would allow us to reject H_0?
- What is the z-score of this $\bar{X}$ value *for the alternative distribution*?
- What is the power of the test?

Solution

- The critical Z value for the test is 1.645
- The $\bar{X}$ with Z-score 1.645 is

$$1.645 = Z_{\overline{X}} = \frac{\overline{X} - \mu_0}{\sigma_{\overline{X}}}$$

Solution

- The critical Z value for the test is 1.645
- The $\bar{X}$ with Z-score 1.645 is

$$1.645 = Z_{\overline{X}} = \frac{\overline{X} - \mu_0}{\sigma_{\overline{X}}}$$

- The Z score of 664.5 *with respect to the alternative hypothesis* is

Solution

- The critical Z value for the test is 1.645
- The $\bar{X}$ with Z-score 1.645 is

$$1.645 = Z_{\overline{X}} = \frac{\overline{X} - \mu_0}{\sigma_{\overline{X}}}$$

- The Z score of 664.5 *with respect to the alternative hypothesis* is
- Power is then

Solution

- The critical Z value for the test is 1.645
- The $\bar{X}$ with Z-score 1.645 is

$$\begin{aligned} 1.645 = Z_{\overline{X}} &= \frac{\overline{X} - \mu_0}{\sigma_{\overline{X}}} \\ \sigma_{\overline{X}} &= \sigma/\sqrt{N} = 1000/\sqrt{100} = 100 \\ \overline{X} &= \mu_0 + (1.645)(100) = 500 + 164.5 = 664.5 \end{aligned}$$

- The Z score of 664.5 *with respect to the alternative hypothesis* is

$$Z_{664.5} = \frac{664.5 - 600}{100} = .645$$

- Power is then

$$Pr(Z > .645) = .2595$$

H_0 is False: The Noncentrality Parameter

$$\begin{aligned} H_0 &: \quad \mu = \mu_0 \\ H_1 &: \quad \mu = \mu_1 \end{aligned}$$

Consider the test statistic $Z = \dfrac{\overline{X} - \mu_0}{\sigma/\sqrt{N}}$ when $X \sim \mathcal{N}(\mu_1, \sigma^2)$.

Noncentrality Parameter

$$
\begin{aligned}
E(Z) &= E\left(\frac{\overline{X}-\mu_0}{\sigma/\sqrt{N}}\right) \\
&= E\left(\frac{\overline{X}-\mu_1+\mu_1-\mu_0}{\sigma/\sqrt{N}}\right)
\end{aligned}
$$

Noncentrality Parameter

$$
\begin{aligned}
E(Z) &= E\left(\frac{\overline{X}-\mu_0}{\sigma/\sqrt{N}}\right) \\
&= E\left(\frac{\overline{X}-\mu_1+\mu_1-\mu_0}{\sigma/\sqrt{N}}\right) \\
&= E\left(\frac{\overline{X}-\mu_1}{\sigma/\sqrt{N}}\right) + E\left(\frac{\mu_1-\mu_0}{\sigma/\sqrt{N}}\right)
\end{aligned}
$$

Noncentrality Parameter

$$
\begin{aligned}
E(Z) &= E\left(\frac{\overline{X}-\mu_0}{\sigma/\sqrt{N}}\right) \\
&= E\left(\frac{\overline{X}-\mu_1+\mu_1-\mu_0}{\sigma/\sqrt{N}}\right) \\
&= E\left(\frac{\overline{X}-\mu_1}{\sigma/\sqrt{N}}\right)+E\left(\frac{\mu_1-\mu_0}{\sigma/\sqrt{N}}\right) \\
&= 0+\frac{\mu_1-\mu_0}{\sigma/\sqrt{N}}
\end{aligned}
$$

Noncentrality Parameter

$$
\begin{aligned}
E(Z) &= E\left(\frac{\overline{X}-\mu_0}{\sigma/\sqrt{N}}\right) \\
&= E\left(\frac{\overline{X}-\mu_1+\mu_1-\mu_0}{\sigma/\sqrt{N}}\right) \\
&= E\left(\frac{\overline{X}-\mu_1}{\sigma/\sqrt{N}}\right) + E\left(\frac{\mu_1-\mu_0}{\sigma/\sqrt{N}}\right) \\
&= 0 + \frac{\mu_1-\mu_0}{\sigma/\sqrt{N}} \\
&= \frac{d}{1/\sqrt{N}} = d\sqrt{N}
\end{aligned}
$$

The noncentrality parameter $\delta = d\sqrt{N}$ is the amount by which the test statistic is shifted away from 0 when the null hypothesis is false.

The shape of the t distribution skews when $\delta \neq 0$ and power is no longer easy to compute by hand.

Using R (z-test)

```
> library('pwrss')
> N <- 100
> d <- 0.10
> ncp <- d*sqrt(N)
> power.z.test(mean=ncp,alpha=0.05,alternative='one.sided')

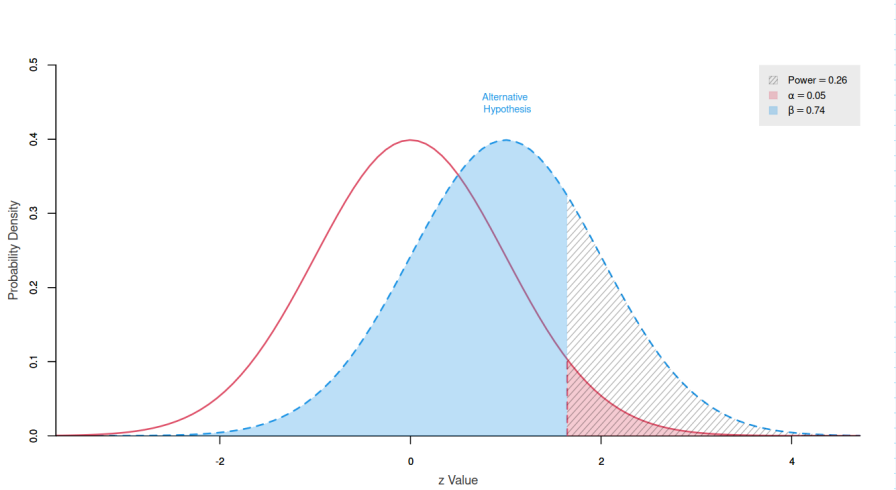
+------------------------------------------------+
|               POWER CALCULATION                |
+------------------------------------------------+

Generic Z-Test

------------------------------------------------
Hypotheses
------------------------------------------------
  H0 (Null Claim) : mean <= null.mean
  H1 (Alt. Claim) : mean > null.mean

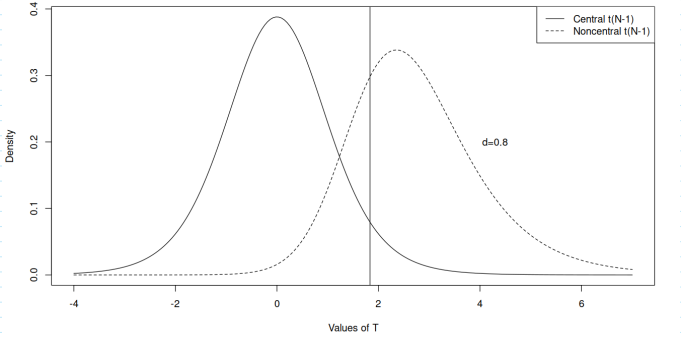
------------------------------------------------
Results
------------------------------------------------
  Type 1 Error (alpha) = 0.050
  Type 2 Error (beta)  = 0.740
  Statistical Power    = 0.26  <<
```

Using R (z-test)



One-Sample t-Test

Problem: the t-distribution shape changes when it is shifted away from 0.
The noncentrality parameter is the same as for the z-test: $d\sqrt{N}$



Katz et al. (1990)

How well do people answer SAT reading comprehension questions when they haven't read the passage to which the questions refer?

Group 1 Read Passages	Group 2 Don't Read Passages
$\overline{X_1} = 69.6$	$\overline{X_2} = 46.6$
$s_1 = 10.6$	$s_2 = 6.8$
$N_1 = 17$	$N_2 = 28$

Effect Size and Noncentrality Parameter

- Effect Size

$$
\begin{aligned}
s_P^2 &= \frac{(17-1)10.6^2 + (28-1)6.8^2}{17+28-2} = 70.84 \\
s_P &= \sqrt{70.84} = 8.42 \\
d &= \left| \frac{69.6 - 46.6}{8.42} \right| = 2.73
\end{aligned}
$$

Effect Size and Noncentrality Parameter

- Effect Size

$$
\begin{aligned}
s_P^2 &= \frac{(17-1)10.6^2 + (28-1)6.8^2}{17+28-2} = 70.84 \\
s_P &= \sqrt{70.84} = 8.42 \\
d &= \left| \frac{69.6 - 46.6}{8.42} \right| = 2.73
\end{aligned}
$$

- Noncentrality Parameter

$$
\begin{aligned}
\delta &= \frac{d}{\sqrt{\frac{1}{N_1} + \frac{1}{N_2}}} \\
&= d\sqrt{\frac{N_1 N_2}{N_1 + N_2}}
\end{aligned}
$$

Two Independent Samples t-test

We only have the sample statistics and not the individual measurements so we have to do the t-test by hand.

```
> N.1 <- 17
> N.2 <- 28
> mean.1 <- 69.6
> mean.2 <- 46.6

> s.1 <- 10.6
> s.2 <- 6.8
> sp <- sqrt((16*10.6^2 + 27*6.8^2)/(17+28-2))
> df <- N.1 + N.2 - 2

> t <- (mean.1 - mean.2)/(sp*sqrt(1/N.1 + 1/N.2))
> 1 - pt(t,df)

1.36e-11
```

Power in R

```
> d <- abs((mean.1 - mean.2)/sp)
> d
[1] 2.732625
> ncp <- d * sqrt((N.1*N.2)/(N.1 + N.2))
> power.t.test(ncp=ncp,df=df,alpha=.05,alternative='one.sided')

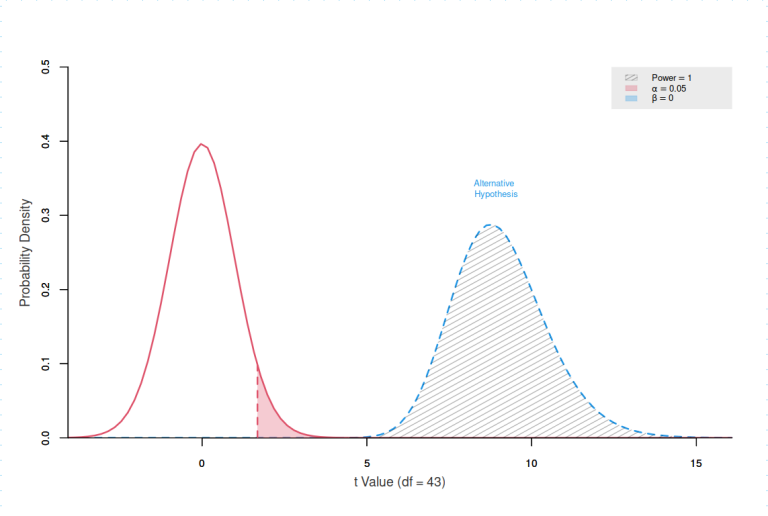
+----------------------------------------------+
|               POWER CALCULATION              |
+----------------------------------------------+

Generic T-Test

------------------------------------------------
Hypotheses
------------------------------------------------
  H0 (Null Claim) : ncp <= null.ncp
  H1 (Alt. Claim) : ncp > null.ncp

------------------------------------------------
Results
------------------------------------------------
  Type 1 Error (alpha)    = 0.050
  Type 2 Error (beta)     = 0.000
  Statistical Power       = 1  <<
```

Power in R


Power = 1
α = 0.05
β = 0
Alternative
Hypothesis
Probability Density
0.0
0.1
0.2
0.3
0.4
0.5
0
5
10
15
t Value (df = 43)

Another Example

Scores on the SAT subtests are normally distributed with mean $\mu = 500$ and standard deviation $\sigma = 100$. Sylvan Learning Systems claimed that their study techniques resulted in a mean increase of 100 points on math subtest. If this were true, how many Sylvan students would you have to test to reject $H_0 : \ \mu = 500$ at an $\alpha = 0.01$ level 95% of the time?

Problem

- Hypotheses:

$$H_0 : \quad \mu = 500$$
$$H_1 : \quad \mu = 600$$

Problem

- Hypotheses:

$$H_0: \quad \mu = 500$$
$$H_1: \quad \mu = 600$$

- The critical Z value for the test is 2.33
- The $\bar{X}$ with Z-score 2.33 satisfies

$$2.33 = Z_{\overline{X}} \quad = \quad \frac{\overline{X} - \mu_0}{\sigma_{\overline{X}}}$$

Sample Size Problem

- The Z score of this $\overline{X}$ *with respect to the alternative hypothesis* is

$$Z_{\overline{X}} = -1.645 \quad = \quad \frac{\overline{X} - 600}{100/\sqrt{N}}$$

Sample Size Problem

- The Z score of this $\overline{X}$ *with respect to the alternative hypothesis* is

$$Z_{\overline{X}} = -1.645 \quad = \quad \frac{\overline{X} - 600}{100/\sqrt{N}}$$

- Solve for N:

$$500 + 2.33(100/\sqrt{N}) \quad = \quad 600 - 1.645(100/\sqrt{N})$$

Sample Size Problem

- The Z score of this $\overline{X}$ *with respect to the alternative hypothesis* is

$$\begin{aligned} Z_{\overline{X}} = -1.645 &= \frac{\overline{X} - 600}{100/\sqrt{N}} \\ \Rightarrow \overline{X} &= 600 - 1.645(100/\sqrt{N}) \end{aligned}$$

- Solve for N:

$$\begin{aligned} 500 + 2.33(100/\sqrt{N}) &= 600 - 1.645(100/\sqrt{N}) \\ 233/\sqrt{N} + 164.5/\sqrt{N} &= 600 - 500 \end{aligned}$$

Sample Size Problem

- The Z score of this $\overline{X}$ *with respect to the alternative hypothesis* is

$$\begin{aligned} Z_{\overline{X}} = -1.645 &= \frac{\overline{X} - 600}{100/\sqrt{N}} \\ \Rightarrow \overline{X} &= 600 - 1.645(100/\sqrt{N}) \end{aligned}$$

- Solve for N:

$$\begin{aligned} 500 + 2.33(100/\sqrt{N}) &= 600 - 1.645(100/\sqrt{N}) \\ 233/\sqrt{N} + 164.5/\sqrt{N} &= 600 - 500 \\ 397.5/\sqrt{N} &= 100 \end{aligned}$$

Sample Size Problem

- The Z score of this $\overline{X}$ *with respect to the alternative hypothesis* is

$$\begin{aligned} Z_{\overline{X}} = -1.645 &= \frac{\overline{X} - 600}{100/\sqrt{N}} \\ \Rightarrow \overline{X} &= 600 - 1.645(100/\sqrt{N}) \end{aligned}$$

- Solve for N:

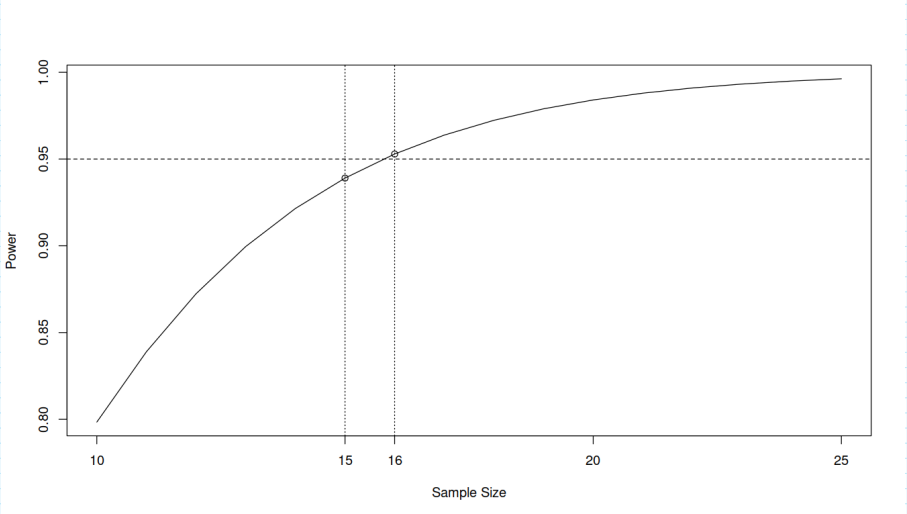
$$\begin{aligned} 500 + 2.33(100/\sqrt{N}) &= 600 - 1.645(100/\sqrt{N}) \\ 233/\sqrt{N} + 164.5/\sqrt{N} &= 600 - 500 \\ 397.5/\sqrt{N} &= 100 \\ \sqrt{N} &= 3.975 \\ N &= 3.975^2 = 15.80 \end{aligned}$$

Sample Size Determination In R

```
> library('pwrss')
> desired.power <- 0.95
> d <- 1.0
> N <- seq(10,25)
> ncp <- d * sqrt(N)
> power <- power.z.test(ncp,alpha=alpha,alternative='one.sided',plot=FALSE)

> first<-which(power>=desired.power)[1]
> plot(N,power,xlab="Sample Size",ylab="Power",type='l')
> points(N[(first-1):first],power[(first-1):first])
> abline(h=desired.power,lty=2)
> abline(v=c(N[first],N[first-1]),lty=3)
> axis(1,at=16,tick=TRUE)
```

Power Curve



Sample Size for z-Test

```
power.z.sample.size <- function(d,alpha,power,
                                test='one.sided') {
  if (test=='one.sided') {
    z_alpha <- qnorm(1-alpha)
  } else if (test=='two.sided') {
    z_alpha <- qnorm(1-alpha/2)
  }
  z_power <- qnorm(power)
  required_n <- ((z_alpha + z_power)/d)^2
  return(ceiling(required_n))
}

> alpha <- 0.01
> desired.power <- 0.95
> d <- 1.0 # effect size to be detected
> power.z.sample.size(d=1,alpha=alpha,test='one.sided',power=desired.power)
[1] 16
```

Two-sample t-Test

```
> alpha <- 0.05
> desired.power <- 0.80
> d <- 0.5

> power.t.student(d,alpha=alpha,alternative='one.sided',
                  power=desired.power,
                  n.ratio=1,var.ratio=1)

+-------------------------------------------------+
|            SAMPLE SIZE CALCULATION              |
+-------------------------------------------------+

Student's T-Test (Independent Samples)

-------------------------------------------------
Hypotheses
-------------------------------------------------
  H0 (Null Claim) : d - null.d <= 0
  H1 (Alt. Claim) : d - null.d > 0

-------------------------------------------------
Results
-------------------------------------------------
  Sample Size            = 51 and 51  <<
  Type 1 Error (alpha)   = 0.050
  Type 2 Error (beta)    = 0.194
  Statistical Power      = 0.806
```

Two-sample t-Test

```
> alpha <- 0.05
> desired.power <- 0.80
> d <- 0.5

> power.t.welch(d,alpha=alpha,alternative='one.sided',
                power=desired.power)

+-----------------------------------------------+
|             SAMPLE SIZE CALCULATION           |
+-----------------------------------------------+

Welch's T-Test (Independent Samples)

-------------------------------------------------
Hypotheses
-------------------------------------------------
  H0 (Null Claim) : d - null.d <= 0
  H1 (Alt. Claim) : d - null.d > 0

-------------------------------------------------
Results
-------------------------------------------------
  Sample Size            = 51 and 51  <<
  Type 1 Error (alpha)   = 0.050
  Type 2 Error (beta)    = 0.194
  Statistical Power      = 0.806
```

Power Curves and Tables

NUMBER OF OBSERVATIONS FOR *t*-TEST OF DIFFERENCE BETWEEN TWO MEANS

Single-sided test / Double-sided test	Level of *t*-test																				
	$\alpha = 0.005$ $\alpha = 0.01$					$\alpha = 0.01$ $\alpha = 0.02$					$\alpha = 0.025$ $\alpha = 0.05$					$\alpha = 0.05$ $\alpha = 0.1$					
$\beta =$	0.01	0.05	0.1	0.2	0.5	0.01	0.05	0.1	0.2	0.5	0.01	0.05	0.1	0.2	0.5	0.01	0.05	0.1	0.2	0.5	
0.05																					0.05
0.10																					0.10
0.15																					0.15
0.20																				137	0.20
0.25															124					88	0.25
0.30										123					87					61	0.30
0.35					110					90					64				102	45	0.35
0.40					85					70				100	50			108	78	35	0.40
0.45				118	68				101	55			105	79	39		108	86	62	28	0.45
0.50				96	55			106	82	45		106	86	64	32		88	70	51	23	0.50
0.55			101	79	46		106	88	68	38		87	71	53	27	112	73	58	42	19	0.55
0.60		101	85	67	39		90	74	58	32	104	74	60	45	23	89	61	49	36	16	0.60
0.65		87	73	57	34	104	77	64	49	27	88	63	51	39	20	76	52	42	30	14	0.65
0.70	100	75	63	50	29	90	66	55	43	24	76	55	44	34	17	66	45	36	26	12	0.70
0.75	88	66	55	44	26	79	58	48	38	21	67	48	39	29	15	57	40	32	23	11	0.75
0.80	77	58	49	39	23	70	51	43	33	19	59	42	34	26	14	50	35	28	21	10	0.80
0.85	69	51	43	35	21	62	46	38	30	17	52	37	31	23	12	45	31	25	18	9	0.85
0.90	62	46	39	31	19	55	41	34	27	15	47	34	27	21	11	40	28	22	16	8	0.90
0.95	55	42	35	28	17	50	37	31	24	14	42	30	25	19	10	36	25	20	15	7	0.95
1.00	50	38	32	26	15	45	33	28	22	13	38	27	23	17	9	33	23	18	14	7	1.00
1.1	42	32	27	22	13	38	28	23	19	11	32	23	19	14	8	27	19	15	12	6	1.1
1.2	36	27	23	18	11	32	24	20	16	9	27	20	16	12	7	23	16	13	10	5	1.2
1.3	31	23	20	16	10	28	21	17	14	8	23	17	14	11	6	20	14	11	9	5	1.3
1.4	27	20	17	14	9	24	18	15	12	8	20	15	12	10	6	17	12	10	8	4	1.4
1.5	24	18	15	13	8	21	16	14	11	7	18	13	11	9	5	15	11	9	7	4	1.5
1.6	21	16	14	11	7	19	14	12	10	6	16	12	10	8	5	14	10	8	6	4	1.6
1.7	19	15	13	10	7	17	13	11	9	6	14	11	9	7	4	12	9	7	6	3	1.7

Value of $\Delta = \frac{\mu_2 - \mu}{\sigma}$

Other Resources

- R pwr package
- R stats package (power.t.test,power.prop.test,power.anova.test)
  - Note that power.t.test in the pwrss package overwrites the power.t.test function in the stats package that is loaded into R by default.
- G*Power (Mac and Windows)